Slide 1:

This is my bachelor thesis, called Patient Zero. I wrote it with Martin Jorgensen, and Jonas Juul supervised it. The question was simple: something spreads through a group of people. Think of a rumour, a virus, or a chain message. Can we look at who ended up affected and work backwards to find the very first person who started it? This presentation is my reflection on that work for this course.

Slide 2:

Going forwards is easy. If you know who started it, you can guess who gets affected next. Going backwards is typically harder. We only see the final picture and thus we don't see who passed it to whom. We just see a web of connections between people. Two details matter. First, that web has no arrows on it, so the direction of spread is hidden. Second, how easily the thing spreads changes everything, so we test many spreading speeds. This matters for tracing a disease, tracking malware, or following fake news.

Slide 3:

Here is what we built, left to right. First: we make two kinds of fake social networks on the computer. Then we let something spread through them at five different speeds. We run this a thousand times for each speed, and we stop every spread once it reaches twenty five people so the cases stay comparable. For each case we measure seven simple clues about the shape of the spread. We feed those clues to a standard prediction model. Finally we compare it against simpler old-school methods, plus pure guessing as a floor.

Slide 4:

The headline result. Our model beats the classic methods almost everywhere. On one network type it climbs from about nine percent correct up to ninety four percent as the spread gets faster. On the other network type it does just as well at the top but loses to the old method when spreading is very slow. The reason is because when something barely spreads, there is almost no trail to follow. Two of the seven clues do about half of the work.

Slide 5:

We trained the model on the made-up data, then pointed it at a real fake-news dataset. It did poorly, around a quarter to a half correct. That looked like failure. But when we trained a brand new model of the exact same type directly on the real data, it got ninety nine percent. The model design was never the problem. The problem was that the practice data did not look like the real world.

Slide 6:

Now I turn three ideas from this course onto my own work. One: Was it a fair contest? I tuned my model carefully but left the older methods at their basic settings. That is an unfair comparison, and it can make my results look better than they are. Two: The gap on real data is not random bad luck, but bias. My fake data assumed everyone behaves the same way, but real people who spread fake news don't. Three: I have to be skeptical of my own findings. One surprising result rested on a very small sample, so it could just be a fluke. I would need a proper statistical check before trusting it.

Slide 7:

Four weaknesses linked to course ideas. We froze every case at twenty five people, which keeps things tidy but may hide real effects. Our model of how things spread is very simple, and real social spreading is messier. We only tried one type of prediction model, so we cannot say a fancier one would not do better. And our simulation had a small ordering quirk that we should test by shuffling.

Slide 8:

Three things I would do next. Train on several different kinds of spreading, not just one, to see if the approach holds up. Repeat the study at different group sizes to separate cause from coincidence. And try a more advanced model to see how much better it could get, while keeping an eye on how much extra computing power it requires.

Slide 9:

My verdict. Finding the original source is possible, but only under the right conditions. It needs enough spread to leave a trail, a helpful network shape, and practice data that matches the real situation. Two wider reflections. This should not be sold as the truth. When it is unsure it is little better than a coin flip, and blaming whoever is most popular online as the source of a rumor would be unfair and a lot of the time wrong. And a more accurate model uses more energy, so clearly accuracy is not worth it at **ANY** cost.